Camera Logic in Three.js

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Reference: Original TexturedCube.jsx Code
https://github.com/humeraaa/hci-cg-textured-cube/blob/main/src/TexturedCube.jsx

Identification of Camera Code in the Existing File (TexturedCube.jsx)

Line 7-15:

This block defines the setCamera() function logic

```
7   function setCamera(width, height) {
8       const camera = new THREE.PerspectiveCamera(
9           75,
10          width / height,
11          0.1,
12          1000
13      );
14      camera.position.z = 5;
15      return camera;
16  }
```

Line 29:

calls the setCamera() function with window.innerWidth and window.innerHeight

```
29      const camera = setCamera(window.innerWidth, window.innerHeight);
```

Line 92:

scene can be rendered from the camera's perspective.

```
92          renderer.render(scene, camera);
```

Separation of Camera Code into a Separate File

TexturedCube.jsx:

Terminal Help Shaheer-154-Camera HCI [Administrator]

TexturedCube.jsx Camera.jsx

src > TexturedCube.jsx > ...

```
import * as THREE from "three";
import { useEffect, useRef } from "react";
import { setCamera } from "./Camera";

const base = import.meta.env.BASE_URL;

function TexturedCube() {

    const mountRef = useRef(null);

    useEffect(() => {

        const width = mountRef.current.clientWidth;
        const height = mountRef.current.clientHeight;

        const scene = new THREE.Scene();

        const renderer = new THREE.WebGLRenderer({ antialias: true, alpha: true });

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER

```
C:\Users\Administrator\Desktop\Shaheer-154-Camera HCI>npm run dev

> three_js_setup@0.0.0 dev
> vite

Port 5173 is in use, trying another one...
Port 5174 is in use, trying another one...

  VITE v5.4.21  ready in 538 ms

  → Local:   http://localhost:5175/hci-cg-textured-cube/
  → Network: use --host to expose
  → press h + enter to show help
```

Camera.jsx:

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TexturedCube.jsx | Camera.jsx ×

src > Camera.jsx > setCamera

```
import * as THREE from "three";

export function setCamera(width, height) {

    const camera = new THREE.PerspectiveCamera(
        75,
        width / height,
        0.1,
        1000
    );

    camera.position.z = 5;

    return camera;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER

```
C:\Users\Administrator\Desktop\Shaheer-154-Camera HCI>npm run dev

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Port 5173 is in use, trying another one...
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  ➜ Local:   http://localhost:5175/hci-cg-textured-cube/
  ➜ Network: use --host to expose
  ➜ press h + enter to show help
```
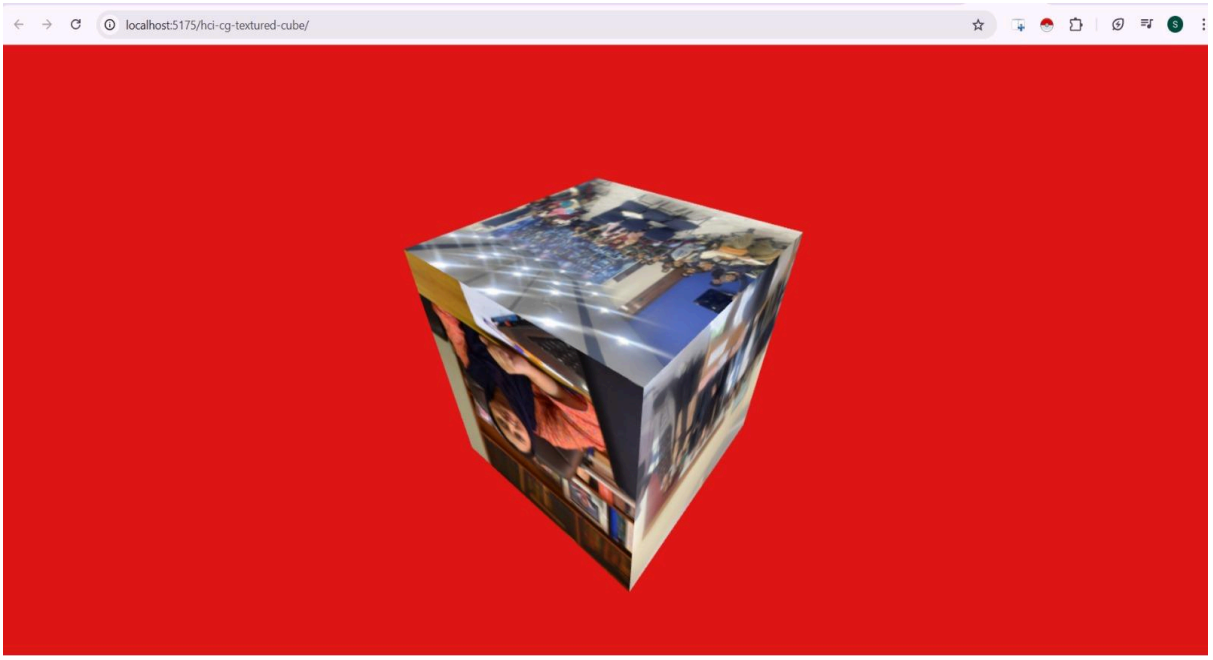

Verification of Output After Code Separation



Refactored Code After Camera Separation (GitHub Repository):
https://github.com/shaheerahmedcoder/HCI-CG-Tasks/tree/main/Camera%20task